Dongwook Kwon

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Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization and Multi-Agent Systems (MAS), with a proven track record of bridging advanced academic research and high-impact industrial applications. Demonstrated research excellence through publications in top-tier natural language processing venues, including an Oral presentation at EMNLP 2025 and a peer-reviewed paper at ACL 2026 focusing on cost-aware LLM cascades. Inventor of multiple registered patents in AI-driven anomaly detection and a key technical lead who has successfully delivered mission-critical autonomous agent systems for global enterprise leaders, including Hyundai Motor Company and LG Electronics. Poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814

Publications

Journal Articles

Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

D. Kwon, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers

R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026. †Equal contribution.

J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. †Equal contribution.

D. Kwon, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

D. Kwon, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Research & Professional Experience

AI Engineer, Intelligent Agent Technology Team — Dec. 2025 – Present
Suresoft Technologies — *Seongnam, Korea*

- Developing an AI validation system utilizing AI agent–based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Research Assistant — Aug. 2021 – Dec. 2025
Bio-Computing & ML Lab, Kwangwoon University — *Seoul, Korea*

- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems).
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

Visiting Researcher Jan. 2025 – Jul. 2025
LG Electronics Canada *Toronto, ON, Canada*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

AI Development Project Participant, Summer Program Jul. 2022 – Aug. 2022
Qualcomm Institute, UC San Diego *San Diego, CA, USA*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University Seoul, Korea
M.S. in Computer Engineering, GPA 4.33 / 4.5 (98.1%) *Mar. 2024 – Feb. 2026*

University of Toronto Toronto, ON, Canada
Graduate Research Program, GPA 3.68 / 4.0 *Jan. 2025 – Jul. 2025*

Kwangwoon University Seoul, Korea
B.S. in Electronics and Communications Engineering, GPA 3.41 / 4.5 (88.1%) *Mar. 2019 – Feb. 2024*

Honors & Awards

Major Awards

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries

Academic Recognition

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering